Computers' Perception and Interpretation of Classical Piano Performances

Research Summary · Ongoing Study

David Eun

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This is a summary of ongoing independent research.
Results are preliminary and subject to change as more analysis is conducted.

Abstract

Although the same classical piano piece is performed, the subjective impressions of listeners differ depending on the pianist. This study tests whether the state-of-the-art audio-language model *Music Flamingo* (Ghosh et al. 2025) can also hear those expressive differences in classical piano performances and presents the findings since August. Using 48 recordings (10 passages across 6 composers with 4–7 performers per passage) and about 1,550 human descriptions from the *Con Espressione* (Cancino-Chacón et al. 2020) and my dataset, I found that the model's descriptions changed between passages but rarely between performers (except for MIDI vs. pianist cases). This was manifest in the performer sensitivity values, or the ratios of same-passage to different-passage semantic distance, which were 0.58 for the model and 0.97 for humans (95% CI gap: 0.07–0.67), but it should be noted that the listeners also agreed weakly with one another. After receiving more access to an A100 GPU, I will observe internal activations, since the differences might be encoded but lost before output.

1 Questions

Because of the different expressive characteristics of pianists (e.g. in timing, dynamics, articulation, and touch), the subjective impressions of listeners depend on the pianist even though they are listening to the same piece.

Do the generated responses of state-of-the-art audio-language models track the expressive differences of performers? If they do, do their descriptions differ the same way as humans' descriptions?

This research project explores these questions.

2 Procedure

The following was the procedure for comparing Music Flamingo's descriptions with human listeners' descriptions of 48 classical piano performance audios over 3 trials.

I compiled a dataset based on the Con Espressione dataset and surveys.

Model: Music Flamingo (`nvidia/music-flamingo-2601-hf`)

GPU: Nvidia A100 80GB

Prompt: "Describe the expressive character of this performance in 3 adjectives."

Recordings: 48 clips across Bach, Beethoven, Brahms, Liszt, Mozart and Schumann, with 4–7 performers per passage, including MIDI performances.

Human data: about 1,550 word-list descriptions.

1. Split each description into terms, lowercase, and strip punctuation.
2. Embed each term with a sentence-embedding model (`all-MiniLM-L6-v2`; Reimers and Gurevych 2019).
3. Calculate the average of the embedded terms within a response, and then the average of the responses for each clip.
4. Compute the distance between every pair of audio clips as $1-\cos(\mathbf{u}, \mathbf{v})$, separately for human and model responses. Label each pair as 'same passage' (different performers) or 'different passage.'

3 Results

Word overlap (model only). Jaccard similarity is shared words / total distinct words, computed on each clip's three descriptors (Table 1). "Expressive" appeared in approximately 90% of outputs.

Table 1: Descriptor overlap between pairs of model outputs.

Clip pair	Mean Jaccard	Identical descriptor sets
Same passage, different pianists	0.60	43%
Different passages	0.33	12%

(Bach MIDI received "playful, whimsical, light-hearted" while all 6 human Bach recordings received "melancholic," "introspective," "contemplative.")

Human vs. model (embeddings). Table 2 summarizes the embedding-based measures.

Table 2: Embedding-based comparisons between human listeners and the model.

Test	What it measures	Human	Model
A. Performer sensitivity	Same-passage / different-passage distance; near 0 = only the passage matters	0.97	0.58
B. Agreement (RSA)	Spearman ρ between human and model distances, same-passage pairs	0.18 (human vs. model)	
C. Human ceiling	Split-half ρ among listeners, Spearman–Brown corrected	0.56	—
E. Individual listeners	Mean distance to single listeners	0.62 (listener–listener)	0.56 (model–listener)

Statistical significance (Test D). The sensitivity gap (A) is significant; human–model agreement (B) is not (Table 3).

Table 3: Significance tests.

Applies to	Method	Result
A. Human − model sensitivity gap	Resample passages with replacement 1,000 times (bootstrap)	95% CI 0.07 to 0.67; excludes 0
B. Human–model agreement	Shuffle performer labels within each passage 1,000 times (permutation)	$p = 0.16$

4 Interpretation

The model appears to track *what* is played more than *how* it is played.

The term "expressive" in approximately 90% of outputs shows the model's small, generic vocabulary when describing pieces. This suggests boilerplate.

Clips of the same passage received identical words 3.6 times as often as clips of different passages (43% vs. 12%). The model's performer sensitivity value (0.58) was significantly below the humans' (0.97).

Human listeners agreed weakly on which performances sounded alike (corrected $\rho = 0.56$). The humans' high performer sensitivity ratio may reflect individual variety instead of shared perception.

One explanation could be that training captions for music rarely distinguish performers, so the model has little reason to encode performer-level differences.

5 Limitations

- **Small sample.** 48 clips and 83 same-passage pairs give wide confidence intervals.
- **Embeddings.** `all-MiniLM-L6-v2` is a general-purpose model, which isn't tuned for musical vocabulary.
- **Vocabulary mismatch.** Humans also described technique ("slow," "clear phrasing"); the model gave only mood adjectives.
- **One model.** Results may not generalize to other audio-language models.
- **Pooling shrinks human distances.** Averaging about 30 answers per clip makes every clip's human vector similar (mean distance 0.07, vs. 0.14–0.24 for the model). The ratios are comparable; absolute distances are not.

6 Future Work

Once I get more access to an A100 GPU, which is required to run inference on Music Flamingo, I plan to observe internal activations, since the differences might be encoded but lost before output.

References

Cancino-Chacón, Carlos Eduardo, Silvan Peter, Shreyan Chowdhury, Anna Aljanaki, and Gerhard Widmer (2020). "On the Characterization of Expressive Performance in Classical Music: First Results of the Con Espressione Game". In: Zenodo. DOI: 10.5281/zenodo.3968828.

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